What Does This Program Do – Senior Division

1. **10-11 C1 What Does This Program Do?**

2. **10-11 C2 What Does This Program Do?**

   What is the final value of S after this program is executed?

```
S = 0
FOR I = 1 TO 12
        FOR J = 1 TO SQRT(I)
                IF I/J = INT(I/J) THEN S=S+1
        NEXT J
NEXT I
PRINT S
END
```

3. **11-12 C1 What Does This Program Do?**

4. **11-12 C3 What Does this Program Do?**

   Given the array at the right, what is the output when this program is executed?

```
for i = 1 to 4
        for j = 1 to 3
                a(i, j)=abs(a(i, j)−a(i, j + 1))
                if a(i, j)/2 = int(a(i, j)/2) then a(i, j)=0 else a(i, j)=a(i, j)+1
                if a(i, j)/4 = int(a(i, j)/4) then a(i, j)=0 else a(i, j)=a(i, j) + 1
                if a(i, j)^.5 = int(a(i, j)^.5) then a(i, j)=0 else a(i, j)=a(i, j)*a(i, j)
        next j
next i
c = 0
for i = 1 to 4
        for j = 1 to 4
                if a(i, j) != 0 then c = c + 1
        next j
next i
print c
end
```

5. **12-13 C1 What Does this Program Do?**

6. **12-13 C3 What Does this Program Do?**

How many of the entries are not 0 after this program is run?

```
dim a(100)
for i = 2 to 25
        a(i)=i
next i
for k = 2 to 25
        if a(k) != 0  then
                for j = 2*k to 25 step k
                        a(j)=0
                next j
        end if
next k
```

7. **13-14 C1 What Does this Program Do?**

8. **13-14 C3 What Does this Program Do?**

How many of the entries are not zero after this program is run?

```
for a = 1 to 4
        for b = 1 to 4
                c(a,b)=a*b-2*a
        next b
next a
for a = 1 to 4
        for b = 1 to 4
                if c(a,b)<0 then c(a,b)=0
                if c(a,b)/2=int(c(a,b)/2) then c(a,b)=c(a,b)/2
                if c(a,b) < 3 then c(a,b) = 0
        next b
next a
end
```

9. **14-15 C1 What Does this Program Do – Loops**

10. 14-15 C3 What Does this Program Do – 2D Arrays

Given array A below, what is printed when this program is run?

```
s = 0
for i = 1 to 4
        for j = 1 to 4
                A(i,j)=A(i,j)+i*j+A(j,i)
        next  j
next i
for i = 1 to 4
        for j = 1 to 4
                if A(i,j)/2!=int(A(i,j)/2) then A(i,j) =A(i,j) + 1
                if A(i,j)/2=int(A(i,j)/2) then A(i,j)=A(i,j)/2
                if A(i,j)>s then s = A(i,j)
        next j
next i
print s
end
```

4	3	9	7
9	2	7	5
1	0	6	8
4	1	2	3

A(1,2) = 1, the bottom row is accessed first

11. 15-16 C1 What Does this Program Do

12. 15-16 C3 What Does this Program Do – Branching

Given array A below, what is printed when this program is run?

```
s = 0
for i = 1 to 4
        for j = 1 to 4
                a(i,j)=a(i,j)-a(j,i)
        next j
next i
for i = 1 to 4
        for j = 1 to 4
                if a(i,j)>= 10 then a(i,j)=a(i,j)-10
                if a(i,j)/2=int(a(i,j)/2) then a(i,j)=a(i,j)/2
                if abs(a(i,j)) < 5 then a(i,j) = 0
        next j
next i
for i = 1 to 4
        for j = 1 to 4
                if a(i,j)=0 then s = s + 1
        next j
next i
print s
end
```

2	6	0	-1
-3	4	-2	3
0	3	-1	-4
8	2	-3	0

A(1,2) = 6

4. **02-03 C4 What Does This Program Do – Strings**

8. **03-04 C4 What Does this Program Do - Strings**

What value is printed when the following program is run?

```
X= "": Y= ""
A = "UNITEDSTATESOFAMERICA"
FOR J = 0 TO LEN [A] − 1
        IF A[J: J] > A[LEN [A] – J − 1: LEN[A] − J − 1] THEN X = X + A[J: J]
NEXT J
FOR K = 0 TO LEN [X] − 1
        IF X[K: K] < "N" THEN Y= Y + X[K: K]
NEXT K
PRINT Y
END
```

12. **04-05 C4 What Does this Program Do – Strings**

What is the final value of A after this code is executed?

```
A = "":  B = "PLEASEEXCUSEMYDEARAUNTSALLY"
FOR I = 0 to LEN [B] – 2
        IF B[I: I] = B[I + 1: I + 1] THEN A = A+ B[I: I]
        IF (B[I: I] = "A" ) OR (B[I: I] = "E" ) THEN A= B[I: I] + A
NEXT I
```